Vyomesh Kumar

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Tampa, FL

SUMMARY

Computer Science student at USF (3.83 GPA, Presidential Scholar) seeking Summer 2027 software engineering and AI/ML internships. Built and deployed full-stack applications using Python, React, FastAPI, REST APIs, LLM integration, OAuth 2.0, and automated testing. Developed a NASA turbofan predictive maintenance pipeline achieving 18.78-cycle MAE across 100 test engines, with a cloud-hosted interface for model inference. Software Lead at USF ACM, developing a voice-controlled AI robot and leading software integration, testing, and debugging.

EDUCATION

University of South Florida | Tampa, FL
B.S. Computer Science | GPA: 3.83/4.0 | Green & Gold Presidential Scholarship | Expected Spring 2028
Relevant Coursework: Computer Organization, Program Design, Programming Concepts, Discrete Structures, Probability & Statistics with Calculus, Linear Systems

PROJECTS

NASA Turbofan Remaining Useful Life Predictor | Python, React, JavaScript, FastAPI, scikit-learn, PyTorch, GitHub Live Demo
- Developed an end-to-end machine learning pipeline on NASA C-MAPSS FD001 sensor data, using engine-level train/validation splits and five-fold grouped cross-validation to prevent engine overlap across evaluation subsets.
- Engineered 88 time-series features, including sensor differences, rolling slopes, and standard deviations; evaluated regression, Random Forest, and PyTorch GRU models, achieving 18.78-cycle MAE and 25.57-cycle RMSE across 100 official test engine endpoints.
- Deployed a responsive React, JavaScript, and CSS frontend integrated with a FastAPI REST API on Render; implemented multipart file uploads, input validation, asynchronous HTTP requests, CORS configuration, and saved-model inference using pandas, NumPy, and joblib.

LinkedIn Career Agent | Python, Claude API, Apify, Streamlit, Pytest, JSON GitHub - Live Demo
- Built and deployed an AI-enabled web application with 10 user-facing features, including profile audits, competitor comparisons, internship eligibility checks, and career roadmap generation; integrated LLM and data APIs with persistent JSON session memory and application logic to maintain conversational context across user interactions.
- Developed 23 pytest test cases using mocked API calls to test input validation, session memory, conversation history, and agent features; identified and fixed a production input-validation bug through testing.

Email Summarizer | Python, Claude API, Gmail API, OAuth2.0, APScheduler GitHub
- Built and deployed production code for an AI-powered email service integrating Gmail and Claude APIs to automate prioritized daily reports summarizing messages from the previous 24 hours, emphasizing reliable automation and user-focused output.
- Designed an automated backend workflow using OAuth authentication and APScheduler; deployed to Railway for reliable 24/7 cloud execution without local intervention, while debugging API integrations and scheduling behavior across external services.

TECHNICAL SKILLS

Languages: Python, C, C++, Java, SQL, JavaScript
Machine Learning: scikit-learn, pandas, NumPy, PyTorch, TensorFlow, XGBoost, Random Forest, Regression, Feature Engineering, Predictive Modeling
Web Development: React, HTML, CSS, FastAPI, Streamlit, REST APIs, OAuth 2.0
Tools & Platforms: Git, GitHub, pytest, MySQL, PostgreSQL, Render, Railway, Google Apps Script, Apify, Notion
Software Engineering: Data Structures & Algorithms, Object-Oriented Programming, API Integration, Full-Stack Development, Software Testing, Version Control, Workflow Automation

LEADERSHIP & ACTIVITIES

Software Lead - ACM Project Hub Team, USF Chapter Aug 2026 – Present
- Developing software for a voice-controlled AI robot using Python, Raspberry Pi, and OpenAI technologies, focusing on voice interaction, AI integration, and movement control; collaborating with the Project Manager and programmers on implementation. Contributing to feature integration, testing, debugging, and team coordination using GitHub and Notion.

Resident Assistant - University of South Florida Housing Aug 2026 – Present
- Support 40+ residents by resolving issues independently, communicating effectively, and adapting to changing priorities.

CERTIFICATIONS

- Machine Learning Specialization (Courses 1 & 2) – Supervised Machine Learning: Regression and Classification; Advanced Learning Algorithms | DeepLearning.AI / Stanford Online (Coursera)
- Advanced Python - LinkedIn Learning